

# Load Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q22-2127

Port Hudson 138 kV  
June 22, 2023

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2127. The Interconnection Customer proposes a 820.9 MW large load (data center) facility interconnecting at the Port Hudson 138 kV substation in the Pennsylvania area. The request was received on July 02, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 715 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$112,606,000.

### 3. Network Impacts

#### 3.1 Load Deliverability Analysis

| Monitored Facility                      | Contingency                             | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------|-----------------------------------------|-------------|--------------|----------|
| DEER VALLEY-NEWLINVILLE 138kV L         | PORT HUDSON-AZTEC 138kV                 | 443.7       | 201          | 100.0    |
| CONNELL-NEWLINVILLE 138kV               | AZTEC-PORT HUDSON 138kV                 | 422.2       | 218          | 100.0    |
| CHIMAYO-DEER VLY 138kV                  | PORT HUDSON-AZTEC 138kV                 | 331.2       | 260          | 100.0    |
| PORT HUDSON-CHIMYO 138kV                | PORT HUDSON-AZTEC 138kV                 | 311.7       | 272          | 100.0    |
| PORT HUDSON-AZTEC 138kV                 | CONNELL-NEWLINVILLE 138kV               | 305.5       | 301          | 100.0    |
| BD CRK-BIG RIV 138kV                    | PORT HUDSON-AZTEC SUBSTATION 138kV Line | 217.9       | 213          | 53.8     |
| BIG RIVER-CONNELL 138kV                 | AZTEC-PORT HUDSON 138kV                 | 195.0       | 226          | 53.8     |
| AZTEC SUBSTATION 345 kV/138 kV TX       | NEWLINVILLE-CONNELL 138kV               | 145.2       | 412          | 77.6     |
| NEVIN-FACE ROCK 138kV L                 | NEEVIN-SAFE HARBOR 138kV                | 142.6       | 228          | 12.3     |
| NEEVIN-PETERSBURG 138kV Line            | STAR PEAK-AZTEC SUBSTATION 345kV        | 140.7       | 332          | 27.3     |
| SWANSEA SUBSTATION-CLAY 138kV           | AZTEC-STAR PEAK 345kV                   | 140.1       | 286          | 27.3     |
| S. MACON-CLAY 138kV L                   | AZTEC-STAR PEAK 345kV L                 | 138.5       | 296          | 27.3     |
| S. MACON-SOLWAY 138kV                   | STAR PEAK-AZTEC 345kV Line              | 135.6       | 316          | 27.3     |
| PETERSBURG-LEADER SUBSTATION 138kV Line | AZTEC-STAR PEAK 345kV Line              | 130.4       | 348          | 27.3     |
| HAMILTON-BAD CREEK 138kV                | TOMBSTONE-CONNELL 345kV                 | 128.2       | 422          | 16.2     |
| HAMILTON-TMBSTONE 138kV L               | CONNELL-TOMBSTONE 345kV                 | 126.0       | 451          | 16.2     |

| Monitored Facility                             | Contingency                     | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------------------|---------------------------------|-------------|--------------|----------|
| SOLWAY-LEANDER 138kV Line                      | AZTEC-STAR PEAK 345kV           | 121.5       | 362          | 27.3     |
| ELIZABETH LAKE-SOUTHPOINT 138kV line           | DENHAM SPRINGS-SAR PEAK 138kV L | 119.2       | 241          | 7.4      |
| NEEVIN-SAFE HARBOR 138kV line                  | NEEVIN-FACE ROCK 138kV line     | 118.2       | 310          | 13.1     |
| STAR PEAK 345 kV/138 kV XFMR                   | TOMBSTONE-BIXBY HILL 500kV      | 115.8       | 477          | 10.7     |
| MILL CTEEK-SOUTHPOINT 138kV                    | DENHAM SPRINGS-STAR PEAK 138kV  | 114.5       | 281          | 7.4      |
| ELIZABETH LAKE-STAR PEAK 138kV L               | STAR PEAK-DNHAM SPRINGS 138kV   | 112.3       | 224          | 7.4      |
| CONNELL 345/138 kV XFMR                        | AZTEC-PORT HUDSON 138kV line    | 110.8       | 434          | 46.2     |
| GRANADA-KINDER MORGAN 138kV Line               | AZTEC-PORT HUDSON 138kV line    | 103.2       | 192          | 5.3      |
| CONOWINGO SUBSTATION 345/138kV Autotransformer | NEWLINVILLE-CONNELL 138kV       | 101.7       | 445          | 5.2      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                               | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------|---------------------------------------------|---------------|-----------------------|
| DEER VALLEY-NEWLINVILLE 138kV          | Rebuild line with higher-capacity conductor | \$1,680,000   | 100% (\$1,680,000)    |
| CONNELL-NEWLINVILLE 138kV              | Rebuild line with higher-capacity conductor | \$3,421,000   | 100% (\$3,421,000)    |
| CHIMAYO-DEER VLY 138kV                 | Rebuild line with higher-capacity conductor | \$4,143,000   | 100% (\$4,143,000)    |
| PORT HUDSON-CHIMYO 138kV               | Rebuild line with higher-capacity conductor | \$2,508,000   | 100% (\$2,508,000)    |
| PORT HUDSON-AZTEC 138kV                | Rebuild line with higher-capacity conductor | \$2,417,000   | 100% (\$2,417,000)    |
| BD CRK-BIG RIV 138kV                   | Rebuild line with higher-capacity conductor | \$1,161,000   | 100% (\$1,161,000)    |
| BIG RIVER-CONNELL 138kV                | Rebuild line with higher-capacity conductor | \$688,000     | 100% (\$688,000)      |
| AZTEC 345-138 kV Transformer           | Replace transformer with higher-rated unit  | \$13,186,000  | 100% (\$13,186,000)   |
| NEEVIN-FACE ROCK SUBSTATION 138kV line | Rebuild line with higher-capacity conductor | \$1,691,000   | 100% (\$1,691,000)    |
| NEEVIN-PETERSBURG 138kV Line           | Rebuild line with higher-capacity conductor | \$3,048,000   | 100% (\$3,048,000)    |
| CLAY-SWANSEA 138kV                     | Rebuild line with higher-capacity conductor | \$2,745,000   | 100% (\$2,745,000)    |
| CLAY-S. MACON 138kV                    | Rebuild line with higher-capacity conductor | \$6,169,000   | 100% (\$6,169,000)    |

| Facility                                       | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| S. MACON-SOLWAY 138kV L                        | Rebuild line with higher-capacity conductor | \$7,631,000   | 100% (\$7,631,000)    |
| LEANDER-PETERSBURG 138kV line                  | Rebuild line with higher-capacity conductor | \$13,652,000  | 100% (\$13,652,000)   |
| HAMILTON-BAD CREEK 138kV                       | Rebuild line with higher-capacity conductor | \$2,337,000   | 100% (\$2,337,000)    |
| HAMILTON-TMBSTONE 138kV L                      | Rebuild line with higher-capacity conductor | \$4,465,000   | 100% (\$4,465,000)    |
| SOLWAY-LEANDER 138kV                           | Rebuild line with higher-capacity conductor | \$2,290,000   | 100% (\$2,290,000)    |
| ELIZABETH LAKE-SOUTHPOINT 138kV line           | Rebuild line with higher-capacity conductor | \$3,073,000   | 54% (\$1,643,000)     |
| SAFE HARBOR-NEEVIN 138kV                       | Rebuild line with higher-capacity conductor | \$4,015,000   | 100% (\$4,015,000)    |
| STAR PEAK 345 kV/138 kV XFMR                   | Replace transformer with higher-rated unit  | \$16,908,000  | 31% (\$5,314,000)     |
| MILL CTEEK-SOUTHPOINT 138kV                    | Rebuild line with higher-capacity conductor | \$5,352,000   | 54% (\$2,861,000)     |
| STAR PEAK SUBSTATION-ELIZABETH LAKE 138kV      | Rebuild line with higher-capacity conductor | \$999,000     | 54% (\$534,000)       |
| CONNELL 345/138 kV XFMR                        | Replace transformer with higher-rated unit  | \$19,515,000  | 100% (\$19,515,000)   |
| GRANADA-KINDER MORGAN 138kV                    | Reconductor line                            | \$1,299,000   | 100% (\$1,299,000)    |
| CONOWINGO SUBSTATION 345/138kV Autotransformer | Replace transformer with higher-rated unit  | \$17,028,000  | 25% (\$4,193,000)     |
|                                                | Total allocated to project                  |               | \$112,606,000         |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.